

An Exact Type D_4 Alternating Normal Form and Condition A for All Garside Powers

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Abstract

Let M be the spherical Artin–Tits monoid of type D_4 , with atoms $\{\sigma'_0, \sigma_0, \sigma_1, \sigma_2\}$, standard Garside element Δ , and ordered covering

$$S_1 = \{\sigma'_0, \sigma_1\}, \quad S_2 = \{\sigma_0, \sigma_1, \sigma_2\}.$$

We compute the complete alternating normal form

$$[\Delta] = (\sigma_2\sigma_0, \sigma_1\sigma'_0, \sigma_0\sigma_1\sigma_2\sigma_1\sigma_0, \sigma'_0\sigma_1\sigma'_0).$$

Its four factors are certified by greatest parabolic right tails in the finite Coxeter group. The leftmost factor then gives $L^*(\Delta, S) = \{\sigma_0, \sigma_2\} = \Phi_\Delta(S_2 \boxminus S_1)$. The criterion of Fromentin–Godelle consequently yields, for every $p, t \in \mathbb{Z}_{\geq 1}$,

$$\text{dpt}(\Delta^{pt}) - 1 = t(\text{dpt}(\Delta^p) - 1).$$

Thus Condition A holds with respect to every Δ^p in this D_4 case. No assertion is made for D_r with $r > 4$ or for Condition B.

1 Statement and context

Write

$$a = \sigma'_0, \quad b = \sigma_0, \quad c = \sigma_1, \quad d = \sigma_2.$$

The Coxeter graph has central vertex c and leaves a, b, d . Thus each leaf satisfies the length-three braid relation with c , while the leaves commute pairwise. For $J \subseteq \{a, b, c, d\}$, let M_J denote the standard parabolic submonoid. If $x \in M$, its right J -tail is the greatest right divisor of x that lies in M_J .

For the ordered covering (S_2, S_1) , the alternating normal form is obtained by taking, successively from the right, the S_1 -tail, the S_2 -tail, and so on. We write it as $[x] = (x_n, \dots, x_1)$, where $x = x_n \cdots x_1$. Its depth is $\text{dpt}(x) = \lfloor n/2 \rfloor$, and $L^*(x, S)$ is the set of atoms that left-divide x_n .

Fromentin and Godelle conjecture in Section 4.3 of [1] the relevant L^* equality for an exceptional even- D_r family. The result below proves precisely its smallest case, $r = 4$. Their Lemma 3.14 supplies one inclusion in the general setting; the missing reverse inclusion here is obtained from an exact normal-form calculation. We make no exhaustive priority claim beyond this comparison with version 1 of their paper.

2 Parabolic tails of a simple lift

Let $W = W(D_4)$ and write $\underline{w}$ for the simple lift of $w \in W$ to M . We use the standard length function ℓ and right weak order; see, for example, [2]. Matsumoto's theorem makes $\underline{w}$ independent of the chosen reduced expression.

Lemma 1. *Suppose J is a set of simple reflections and*

$$w = w^J w_J, \quad w_J \in W_J, \quad \ell(w) = \ell(w^J) + \ell(w_J),$$

where w^J has no right descent in J . Then the right J -tail of the simple element $\underline{w}$ is $\underline{w}_J$.

Proof. The length-additive factorization lifts to $\underline{w} = \underline{w}^J \underline{w}_J$, so $\underline{w}_J$ is a parabolic right divisor. Every right divisor of a simple element is simple. Hence any parabolic right divisor is $\underline{v}$ for some $v \in W_J$ with $v \leq_R w$. The standard parabolic decomposition, characterized by the absence of right descents of w^J in J , makes w_J the greatest such element in right weak order. Thus $v \leq_R w_J$, so $\underline{v}$ right-divides $\underline{w}_J$. This proves greatestness. $\square$

We now give the exact Coxeter calculation. Work in $\mathbb{R}^4$ with simple roots

$$\alpha_b = e_1 - e_2, \quad \alpha_c = e_2 - e_3, \quad \alpha_d = e_3 - e_4, \quad \alpha_a = e_3 + e_4.$$

Thus b, c, d swap respectively the coordinate pairs $(1, 2)$, $(2, 3)$, and $(3, 4)$, while $a(e_3) = -e_4$, $a(e_4) = -e_3$, and a fixes e_1, e_2 . The positive roots are $e_i \pm e_j$ for $i < j$, twelve in total.

The source representative of the standard Garside element is

$$bacbacdcbacd.$$

Direct multiplication of the signed-permutation matrices gives

$$bacbacdcbacd = -I_4 = dbcabcdcbaca. \quad (2.1)$$

Since $-I_4$ sends all twelve positive roots to negative roots, it has Coxeter length 12. Both words in (2.1) have twelve letters and are therefore reduced. Matsumoto's theorem identifies their positive lifts, so the second word represents Δ in M .

Starting at the right of that word gives the following length-additive factorizations. The last column records the complete numbers of parabolic right suffixes; these counts are not needed by Lemma 1, but provide an additional finite check of greatestness.

step	J	before	residual	tail	suffixes
1	$\{a, c\}$	$dbcabdcdbaca$	$dbcabdcdb$	aca	6
2	$\{b, c, d\}$	$dbcabdcdb$	$dbca$	$bcdcb$	12
3	$\{a, c\}$	$dbca$	db	ca	3
4	$\{b, c, d\}$	db	1	db	4

The corresponding length equalities are, in order,

$$12 = 9 + 3, \quad 9 = 4 + 5, \quad 4 = 2 + 2, \quad 2 = 0 + 2. \quad (2.2)$$

For completeness, the absence of a right descent in each indicated parabolic can be read directly from the following images of its simple roots under the residual:

$$\begin{aligned} dbcabdcdb : \quad & \alpha_a \mapsto e_2 - e_3, \quad \alpha_c \mapsto e_3 + e_4; \\ dbca : \quad & \alpha_b \mapsto e_2 - e_4, \quad \alpha_c \mapsto e_3 + e_4, \quad \alpha_d \mapsto e_1 - e_3; \\ db : \quad & \alpha_a \mapsto e_3 + e_4, \quad \alpha_c \mapsto e_1 - e_4. \end{aligned}$$

All displayed roots are positive; the residual in the fourth row is the identity. Each tail uses only letters in its displayed J . Equations (2.2) and Lemma 1 therefore prove that all four tails are the greatest required parabolic right divisors.

3 The equality and the all-powers consequence

Theorem 2. *For the covering (S_2, S_1) in the type D_4 Artin–Tits monoid,*

$$[\Delta] = (db, ca, bcdbc, aca).$$

Consequently,

$$L^*(\Delta, S) = \{b, d\} = \{\sigma_0, \sigma_2\} = \Phi_\Delta(S_2 \boxminus S_1).$$

Proof. Reading the four tails above from left to right gives the stated alternating normal form. Its leftmost factor is $db = bd$. Exact Coxeter lengths give

$$\ell(bdb) = \ell(ddb) = 1, \quad \ell(adb) = \ell(cdb) = 3,$$

whereas $\ell(db) = 2$. Hence the left descents of db are exactly $\{b, d\}$, which is the asserted L^* set.

It remains to identify the right-hand side. No atom commutes with both a and c : the atoms a, c fail to commute with one another, while each of b, d fails to commute with c . Thus $S_1^\perp = \emptyset$ and

$$S_2 \boxminus S_1 = S_2 \setminus (S_1 \cup S_1^\perp) = \{b, d\}.$$

Moreover the longest element of $W(D_4)$ is $-I_4$, so conjugation by it fixes every simple reflection. The associated Garside automorphism Φ_Δ is therefore the identity. This proves the equality. $\square$

Corollary 3. *For every pair $p, t \in \mathbb{Z}_{\geq 1}$,*

$$\text{dpt}(\Delta^{pt}) - 1 = t(\text{dpt}(\Delta^p) - 1).$$

Equivalently, the covering satisfies Condition A with respect to every Δ^p .

Proof. Both induced parabolic graphs are connected, the covering is proper, and centrality of Δ makes the covering both Δ -regular and Δ -fixed in the terminology of [1]. Theorem 2 supplies the L^* equality required by their Corollary 4.7. That criterion gives Condition A for the base Δ :

$$\text{dpt}(\Delta^m) - 1 = m(\text{dpt}(\Delta) - 1) \quad (m \in \mathbb{Z}_{\geq 1}). \quad (3.1)$$

Apply (3.1) first with $m = pt$ and then with $m = p$. This yields

$$\text{dpt}(\Delta^{pt}) - 1 = pt(\text{dpt}(\Delta) - 1) = t(\text{dpt}(\Delta^p) - 1),$$

including the endpoint cases $p = 1$ and $t = 1$. No finite-power extrapolation is involved. $\square$

4 Verification and scope

The matrix products, lengths, parabolic decompositions, suffix counts, and left descents above were checked by a companion Python program using only integer 4×4 matrices and exhaustive enumeration. It enumerates all 192 elements of $W(D_4)$ and the relevant parabolic groups of orders 6 and 24. For each of the four rows it enumerates every parabolic right suffix and verifies that each

right-divides the displayed tail. There is no floating-point, randomized, or finite-power inference in the argument.

As a methodological comparison only, Zeng [3] studies an explicit two-generation problem for an equal-sign fiber product of finite symmetric groups. That separate work illustrates exact finite-permutation-group certification, but none of its conclusions implies an Artin–Tits normal-form identity or Condition A and none is used in the proof above.

The present calculation settles only the stated D_4 member of the family identified in [1]. It neither treats D_r for $r > 4$ nor addresses Condition B or a left-orderability conclusion.

Computational and AI assistance disclosure. Exact computations were performed with the deterministic Python verifier described above. A generative AI system assisted with organizing the proof, checking the exposition against the accepted exact certificate, and preparing the L^AT_EX source. All claims used in the proof are stated in the article and are reproducible from the displayed signed-permutation model; the author takes responsibility for the manuscript.

References

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